

Yifan Li

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Education

University of Connecticut

PHD IN STATISTICS

Connecticut, USA

Sep 2018 - Sep 2023

University of Wisconsin - Madison

MASTER IN STATISTICS

Wisconsin, USA

Sep 2016 - May 2018

Nanjing University

BACHELOR IN STATISTICS

Jiangsu, China

Sep 2013 - Jun 2017

Work Experience

Quantitative Trading Book in Ernst & Young U.S. LLP

New York, USA

SENIOR CONSULTANT

Oct 2023 - Present

- **Generative AI for Ongoing Performance Assessment (OPA) Reporting**
 - Developed a multi-agent Generative AI system using **LangGraph** to automate OPA validation report generation for trading book models.
 - Orchestrated specialized AI agents responsible for data cleaning, trade pricing, result validation, sensitivity testing, and summarization.
 - Enabled agents to dynamically select analytical tools from third-party **MCP libraries**.
 - Automatically generated end-to-end OPA reports, improving consistency and review efficiency for model risk management teams.
- **Credit Card Fraud Detection using Unsupervised Learning**
 - Developed an unsupervised fraud detection system from high-dimensional, anonymized consumption data.
 - Implemented an **Isolation Forest** algorithm to address the absence of labels, isolating anomalies by randomly partitioning features and measuring tree path lengths.
 - Optimized for **low computational latency**, ensuring high-speed processing on hardware-constrained environments.
 - Navigated compliance restrictions where all variables were **PCA-transformed or masked**, delivering robust performance without relying on explicit feature semantics.
- **CCAR Stress Testing for Trading Book (ETD & OTC)**
 - Designed and implemented a CCAR stress testing framework to quantify P&L impacts for derivatives under regulatory stress scenarios.
 - Built trade-specific object-oriented classes and leveraged a **Depth-First Search (DFS)** traversal to efficiently handle heterogeneous transaction structures stored as complex nested dictionaries.
 - Produced transparent and intuitive stress attribution reports, decomposing total P&L into individual stress factor impacts for senior risk stakeholders.
- **Credit Card Approval with Unbalanced Data and Outliers**
 - Decide who to approve or decline for credit based on historical repayment records.
 - Adding new missing indicator variables before applying imputing missing value after checking randomness.
 - Generate features based on the distribution of outliers and assign different weights on unbalanced responses.
 - Fit **logistic regression**, **XGBoost**, and **Random Forest** models separately and use the linear combination of three models as final model.
- **Value-at-Risk (VaR) Risk Matrix Construction and Maintenance**
 - Maintained and enhanced a VaR calculation framework by regularly updating risk matrices used across the trading book.
- **Optimization of American Options Pricing**
 - Applied the **American Monte Carlo (AMC)** method to replace a nested Monte Carlo approach for pricing American-style options.
 - Reduced computational complexity from **$O(n^2)$ to $O(n)$** , achieving substantial runtime improvements while preserving pricing accuracy.

Bank of China International Holdings Limited

Shanghai, China

SECURITIES ANALYST ASSISTANT (INTERN)

Jun 2021-Sep 2021

- Focused on battery and new energy industry. Predicted the short- and long-term performance of stocks of related companies based on time series model with a spike-and-slab error.
- Adjusted the prediction under a multinomial model based on the performance of correlated companies and avoided making an over-optimistic forecast compared with previous model.

HUATAI SECURITIES CO., LTD. (HTSC)

Jiangsu, China

DATA ANALYST (INTERN)

Jul 2017-Sep 2017

- Unsupervised screened visitors with a strong desire to buy products based on their records on company's APP.
- Cleaned and reshaped the 17 million visitor records by summarizing operations from the same visitor.
- Extracted useful variables by PCA (principal component analysis) method.
- Divided visitors into five groups by K-means methods and assigned visitors labels by their group.
- Fitted a decision tree with labeled data which could tag new visitor within 20 seconds while the target is 1 min.

Skill

- **Language:** Mandarin Chinese (Native), English
- **Coding/Database Languages:** Master R, Python, GitHub, Latex, Nimble, JUGS and HPC, familiar with SQL, SAS, MATLAB, C++ and Julia.
- **Certificate:** CFA level 1